

# Applying Exponent Laws

---

## MA.8.NSO.1.3

I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

# Warm-Up

---

1. Find the value:  $2^{-3}$
2. Rewrite with a positive exponent:  $\frac{1}{7^{-2}}$
3. Which is larger,  $4^3$  or  $3^4$ ?
4. Write the reciprocal of  $-\frac{2}{5}$

# Warm-Up

---

1. Find the value:  $2^{-3}$   $\frac{1}{8}$
2. Rewrite with a positive exponent:  $\frac{1}{7^{-2}}$   $7^2$
3. Which is larger,  $4^3$  or  $3^4$ ?  $3^4$  ( **$81 > 64$** )
4. Write the reciprocal of  $-\frac{2}{5}$   $-\frac{5}{2}$

# Notes

---

## I. The pattern with a fraction base

---

| Expression   | Expanded        | Value  |
|--------------|-----------------|--------|
| $(2/5)^2$    | $2/5 \cdot 2/5$ | $4/25$ |
| $(2/5)^1$    | $2/5$           | $2/5$  |
| $(2/5)^0$    | $1$             | $1$    |
| $(2/5)^{-1}$ | $5/2$           | $5/2$  |
| $(2/5)^{-2}$ | $5/2 \cdot 5/2$ | $25/4$ |

# Notes

---

## II. Flip the base, make the exponent positive

---

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \quad (a, b \neq 0)$$

$$\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4} \qquad \left(-\frac{2}{3}\right)^{-5} = \left(-\frac{3}{2}\right)^5$$

**The sign stays with the fraction. Flip the numbers, not the sign.**

A whole number is a fraction over 1:  $4^{-3} = \left(\frac{4}{1}\right)^{-3} = \left(\frac{1}{4}\right)^3$

# Notes

---

## III. Several laws at once — the plan is still yours

---

$$\left(\frac{2.5^5}{2.5^3}\right)^{-1} = (2.5^2)^{-1} = 2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

Any correct order of laws gives the same value. Finish with a positive exponent.

# Example 1

---

$$\left(-\frac{7}{8}\right)^{-2}$$

**Rewrite with a positive exponent, then find the value.**

# Example 1

---

$$\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^2$$

$$\left(-\frac{8}{7}\right)^2 = \frac{64}{49}$$

**Answer: 64/49**

# Your Turn

---

$$\left(-\frac{5}{2}\right)^{-3}$$

**Rewrite with a positive exponent, then find the value.**

# Your Turn

---

$$\left(-\frac{5}{2}\right)^{-3}$$

**Rewrite with a positive exponent, then find the value.**

Flip the fraction:  $\left(-\frac{5}{2}\right)^{-3} = \left(-\frac{2}{5}\right)^3$

$$\left(-\frac{2}{5}\right)^3 = -\frac{8}{125}$$

**Answer:**  $-\frac{8}{125}$

## Example 2

---

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4$$

**Write an equivalent expression with one base and a positive exponent.**

## Example 2

---

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4 = \left(\frac{5}{8}\right)^{-2+4} = \left(\frac{5}{8}\right)^2$$

$$\left(\frac{5}{8}\right)^2 = \frac{25}{64}$$

**Answer:  $(5/8)^2 = 25/64$**

# Your Turn

---

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

**Write an equivalent expression with one base and a positive exponent.**

# Your Turn

---

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

**Write an equivalent expression with one base and a positive exponent.**

Add the exponents, then flip the fraction:

$$\left(-\frac{5}{4}\right)^{-7+(-3)} = \left(-\frac{5}{4}\right)^{-10} = \left(-\frac{4}{5}\right)^{10}$$

**Answer:**  $\left(-\frac{4}{5}\right)^{10}$

# Whiteboards • Question 1 of 9

---

A copier is set to shrink a page to  $\frac{2}{3}$  of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

**What fraction of the original page's size is the last copy?**

**Answer it.**

# Whiteboards • Question 1 of 9

---

A copier is set to shrink a page to  $\frac{2}{3}$  of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

**What fraction of the original page's size is the last copy?**

Four copies:  $\left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4} = \frac{16}{81}$

**Answer:  $\frac{16}{81}$**

# Whiteboards • Question 2 of 9

---

$$\left(\frac{3}{-5}\right)^{-9}$$

**Answer it.**

# Whiteboards • Question 2 of 9

---

$$\left(\frac{3}{-5}\right)^{-9}$$

Flip the fraction:  $\left(\frac{3}{-5}\right)^{-9} = \left(\frac{-5}{3}\right)^9$

**Answer:**  $\left(\frac{-5}{3}\right)^9$

# Whiteboards • Question 3 of 9

---

$$\left(-\frac{8}{9}\right)^{-2}$$

**Answer it.**

# Whiteboards • Question 3 of 9

---

$$\left(-\frac{8}{9}\right)^{-2}$$

Flip the fraction:  $\left(-\frac{8}{9}\right)^{-2} = \left(-\frac{9}{8}\right)^2$

$$\left(-\frac{9}{8}\right)^2 = \frac{81}{64}$$

**Answer:**  $\frac{81}{64}$

# Whiteboards • Question 4 of 9

---

$$\left(\frac{21}{4}\right)^{-5}$$

**Answer it.**

# Whiteboards • Question 4 of 9

---

$$\left(\frac{21}{4}\right)^{-5}$$

Flip the fraction:  $\left(\frac{21}{4}\right)^{-5} = \left(\frac{4}{21}\right)^5$

**Answer:**  $\left(\frac{4}{21}\right)^5$

# Whiteboards • Question 5 of 9

---

$$\left(-\frac{7}{10}\right)^{-3}$$

Which expression is equivalent?

**A.**  $(7/10)^3$

**B.**  $(-10/7)^3$

**C.**  $(-7/10)^3$

**D.**  $(10/7)^3$

**Answer it.**

# Whiteboards • Question 5 of 9

---

$$\left(-\frac{7}{10}\right)^{-3}$$

Which expression is equivalent?

Flip the fraction. The sign stays:  $\left(-\frac{7}{10}\right)^{-3} = \left(-\frac{10}{7}\right)^3$

**Answer: B —  $(-10/7)^3$**

# Whiteboards • Question 6 of 9

---

$$\left(\frac{2}{9}\right)^{-3} \div \left(\frac{2}{9}\right)^{-7}$$

Which expression is equivalent?

**A.**  $(9/2)^4$

**B.**  $(2/9)^{-10}$

**C.**  $(2/9)^4$

**D.**  $(2/9)^{21}$

**Answer it.**

# Whiteboards • Question 6 of 9

---

$$\left(\frac{2}{9}\right)^{-3} \div \left(\frac{2}{9}\right)^{-7}$$

Which expression is equivalent?

$$\left(\frac{2}{9}\right)^{-3} \div \left(\frac{2}{9}\right)^{-7} = \left(\frac{2}{9}\right)^{-3 - (-7)} = \left(\frac{2}{9}\right)^4$$

**Answer: C —  $(2/9)^4$**

# Whiteboards • Question 7 of 9

---

$$\left(\frac{3}{8}\right)^{-1} \cdot \left(\frac{8}{3}\right)^1$$

**Answer it.**

# Whiteboards • Question 7 of 9

---

$$\left(\frac{3}{8}\right)^{-1} \cdot \left(\frac{8}{3}\right)^1$$

$$\left(\frac{3}{8}\right)^{-1} = \left(\frac{8}{3}\right)^1$$

$$\left(\frac{8}{3}\right)^1 \cdot \left(\frac{8}{3}\right)^1 = \left(\frac{8}{3}\right)^2 = \frac{64}{9}$$

**Answer:**  $\frac{64}{9}$

# Whiteboards • Question 8 of 9

---

$$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$$

**Answer it.**

# Whiteboards • Question 8 of 9

---

$$\left[ \left( \frac{2}{3} \right)^5 \cdot \left( \frac{2}{3} \right)^{-2} \right]^3$$

Inside:  $\left( \frac{2}{3} \right)^5 \cdot \left( \frac{2}{3} \right)^{-2} = \left( \frac{2}{3} \right)^{5 + (-2)} = \left( \frac{2}{3} \right)^3$

$$\left( \left( \frac{2}{3} \right)^3 \right)^3 = \left( \frac{2}{3} \right)^{3 \cdot 3} = \left( \frac{2}{3} \right)^9$$

**Answer:**  $\left( \frac{2}{3} \right)^9$

## Whiteboards • Question 9 of 9

---

Marco says  $\left(-\frac{4}{11}\right)^{-7}$  can be rewritten as  $\left(\frac{4}{11}\right)^7$

because 'two negatives make a positive.' Is he right?

**Write your answer in sentences.**

# Whiteboards • Question 9 of 9

---

Marco says  $\left(-\frac{4}{11}\right)^{-7}$  can be rewritten as  $\left(\frac{4}{11}\right)^7$

because 'two negatives make a positive.' Is he right?

The negative exponent flips the fraction. The sign stays.

$$\left(-\frac{4}{11}\right)^{-7} = \left(-\frac{11}{4}\right)^7$$

**Answer: No —  $(-4/11)^{-7} = (-11/4)^7$**

# Independent Set

---

1. Rewrite with a positive exponent.  $\left(\frac{3}{4}\right)^{-2}$
2. Find the value.  $\left(\frac{6}{7}\right)^{-2}$
3. Rewrite with a positive exponent.  $\left(-\frac{5}{6}\right)^{-3}$
4. Find the value.  $\left(\frac{1}{2}\right)^{-4}$
5. Find the value.  $\left(\frac{3}{2}\right)^{-1} \cdot \left(\frac{3}{2}\right)^3$
6. Find the value of  $n$ .  $\left(\frac{4}{7}\right)^n = \left(\frac{7}{4}\right)^3$

# IXL

---

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Identify equivalent expressions involving exponents I — VLM